Tanishq Saxena

2nd Year Undergraduate
Department of Computer Science and Engineering
Indian Institute of Technology Kharagpur

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Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
July 2024 - Present	4-year B.Tech course in Computer Science and Engineering	Indian Institute of Technology, Kharagpur	9.43/10
2024	Class XII (CBSE)	Saarthi International School, Delhi	95.4%
2022	Class X (CBSE)	VBPS, Noida	98.6%

Internships & Projects

Local INN: Localization using Invertible Neural Networks | Python, PyTorch | Group Project *Dec 2025*

- Demonstrated localization accuracy comparable to particle filters while maintaining stable error at high speeds (5 m/s), using a VAE-based INN that outputs full pose distributions with uncertainty (covariance) estimation.
- Reduced localization latency from 45 Hz (particle filter) to 270 Hz via an INN pipeline, enabling reliable high-speed navigation.

Civix | React.js, Tailwind CSS, Node.js, PostgreSQL | SIH Group Project 2025 *Aug 2025 - Sept 2025*

- Built backend services for a full-stack civic issue reporting platform with JWT authentication, role-based access control, and a PostgreSQL-backed issue lifecycle (Open → In Progress → Resolved).
- Created normalized schemas and REST APIs, and integrated map-based issue visualization to keep queries performant and user views responsive at city-scale usage.

Target-Biased Obstacle Avoidance | Python, NumPy *Mar 2025*

- Designed an autonomous navigation algorithm using LIDAR data for obstacle avoidance toward specified waypoints.
- Enhanced the Follow-the-Gap algorithm with a custom potential function balancing safety, gap width, and target direction.
- Engineered efficient sensor preprocessing and decision logic for robust performance in dynamic environments.

Sparse Optical Flow | Python *Mar 2025*

- Implemented the pyramidal Lucas–Kanade optical flow algorithm for sparse motion tracking between frames.
- Prepared image pyramids, tracking logic, and validation pipelines emphasizing numerical stability.

Course Work Information

Linear Algebra, Numerical and Complex Analysis | Probability and Statistics | Discrete Structures | Formal Language and Automata Theory | Algorithms-1 | Systems Programming | Advanced Calculus | Software Engineering | Completed a 3-course Machine Learning Specialization, covering supervised, unsupervised, and reinforcement learning

Experience

Undergraduate Researcher | Autonomous Ground Vehicles (AGV) | IIT Kharagpur *Apr 2025 - Present*

- Selected for the Software/AI team via project-based evaluation; contributed to SLAM, localization, and ROS-based navigation using sparse optical flow.

Associate Member | Kharagpur Blockchain Society | IIT Kharagpur *Dec 2024 - Oct 2025*

Member | CodeClub | Department of Computer Science and Engineering | IIT Kharagpur *Mar'25 - Present*

Skills and Expertise

Competitive Programming: Specialist on Codeforces noobhacker123
Programming Languages: Python | JavaScript | C | C++ | HTML | CSS | SQL | Prolog | Bash
Frameworks/Libraries: Django | Node.js | Express | ROS2 | NumPy | Bootstrap
Tools, Databases & Technologies: SQL | Git | Docker | GitHub | Postman | Linux | Burp Suite | REST APIs
Security & CTF Skills: Web Exploitation (XSS, SQLi, CSRF) | HTTP & Session Security | Cross-Site Scripting (XSS)

Awards and Achievements

- **AIR 341** in JEE Advanced 2024 among 1,80,200 candidates, securing admission to IIT Kharagpur
- **AIR 1152** in JEE Mains 2024 among 14,15,110 candidates nationwide
- Qualified for **IICPC Codefest 2026 Regionals** with an **All India Rank 392** among 13,000+ participants
- **State Top 1%** in National Standard Examination in Chemistry (**NSEC 2023**)
- Represented **IIT Kharagpur** at the **4th RoboRacer Sim League**, ranked **7th globally** in finals and **5th** in qualifiers
- **CTF Ranking: 71st** (NiteCTF) as a member of **Team libbabel** (Library of Babel, Cybersecurity Team of IIT Kharagpur)